

Unreal Horde Server Deployment (Windows)



Hidde Derks

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Step 1.

Step 2.

Step 3.

Step 4.

Step 5.

Step 6.

Step 7.

Step 8.

Initially created for the Breda University of Applied Sciences Master's of Game Technologies

To start benefiting from the Unreal Build Accelerator you need to have a server set up for agents to connect to. In case you're looking for agent set up, choose one of the following:

Horde Agent (Windows)

Horde Agent (Docker/Linux)

This guide will focus on deploying the Horde Server in Windows, if you're looking for the Docker deployment, go here:

Horde Server (Docker)

Before getting started with this guide, you do need to have access to Unreal Engine's source code on GitHub, guide for which can be found here: <https://www.unrealengine.com/en-US/ue-on-github>. And a machine that will become your Horde Server.

Step 1.

Head to the [releases tab of the Unreal GitHub page](#), and download the most recent `UnrealHordeServer.msi` (at time of writing it's under the Unreal Engine 5.7.0 release).

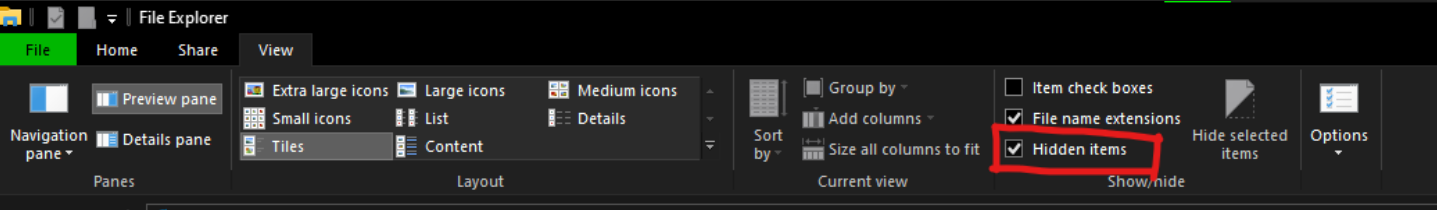
Step 2.

Run the `UnrealHordeServer.msi` installer with default settings, potentially moving the install directory if desired. Other settings can be tweaked in future.

At the end of the installation, it will open the landing page in your browser.

Step 3.

For this step, we need to enable hidden files in file explorer, as `ProgramData` by default is a hidden folder in windows:



We'll probably want to tweak the `server.json` found in the data directory (`C:/ProgramData/Epic/Horde/Server/server.json` by default). A more complete configuration is provided below (which will need some final tweaking based on the desired setup), so copy the contents into your `server.json` . In this configuration some parts are included that will not be required in a default setup, and therefore commented out, but are included as easy starting points for further configuration (e.g. TLS certificates for HTTPS support):

[Tweak the file according to your requirements \(see comments for details\)](#)

```
// Changes to this file requires a restart of the server
{
  "Horde": {

    // How to authorize users to log in to the server. Auth is disabled
    // by Horde itself ("Horde") or to use an external OIDC auth provider
    "AuthMethod": "Horde",

    // Url to the server, needs to match the server host protocol with h
    "ServerUrl": "http://horde.example.com",

    // Path to read the global Horde configuration from. This can be a p
    // The specified file is continuously monitored for updates and will
    "ConfigPath": "globals.json",

    // List of Perforce servers for use specifically by the server (eg.
    // Configuration for the server may be stored in a more secure locat
    "Perforce": [
      {
        "id": "SomePerforceServer",
        "serverAndPort": "ssl:perforce.example.com:1666",
        "credentials": {
          "userName": "Horde",
          "password": "password"
          //"ticket": "ticket (optional)"
        }
      }
    ],
    "plugins": {
      //      "build": {
      //        "enabled": "true",
      //        "perforce": [
      //          {
      //            "id": "SomePerforceServer",
      //            "serverAndPort": "ssl:perforce.example.com:1666",
      //            "credentials": {
      //              "userName": "Horde",
      //              "password": "password"
      //              //"ticket": "ticket (optional)"
      //            }
      //          }
      //        ]
      //      },
      "compute": {
        "enabled": "true",
      },
      "storage": {
        "backends": [
          {
            "enabled": "true",
            "id": "HostStorage",
            "type": "FileSystem",
            "baseDir": "C:/ProgramData/Epic/Horde/Server/Storage"
          }
        ]
      },
      "analytics": {
        "enabled": "true",
        "sinks": {
          "mongo": {
            "enabled": "true",

```

```

        "retainDays": "90"
    }
}
},
{
    // Configure an editor telemetry sink that stores data in the bundle
    // This stores raw telemetry data; aggregated metrics can be computed
    // needing to scan this data, and is expired separately.
    "Telemetry": [
        {
            "Type": "Mongo",
            "RetainDays": 3
        }
    ],
    // Whether to use the local Perforce environment for reading configuration
    "UseLocalPerforceEnv": false
},
// TLS certificate configuration. AllowInvalid is required if using a self-signed cert
// "Kestrel": {
//     "Certificates": {
//         "Default": {
//             "Path": "C:/Path/To/Horde/Certs/Horde.crt",
//             "KeyPath": "c:/Path/To/Horde/Certs/Horde.key",
//             "AllowInvalid": "true"
//         }
//     }
// }
}

```

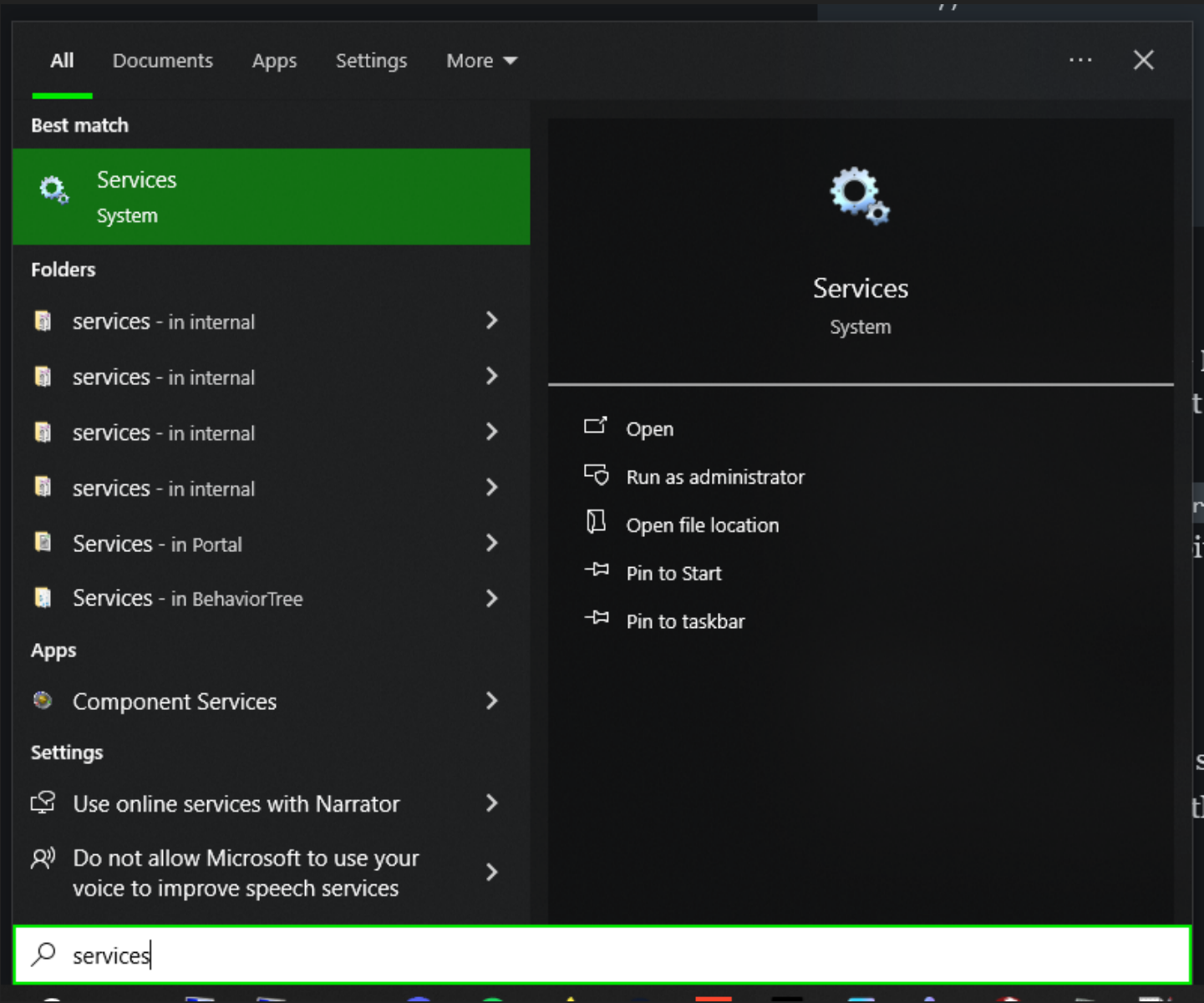
server.json

Should you want to alter the ports configured in the installer, those would have to be tweaked in the Registry at `Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Epic Games\Horde\Server`

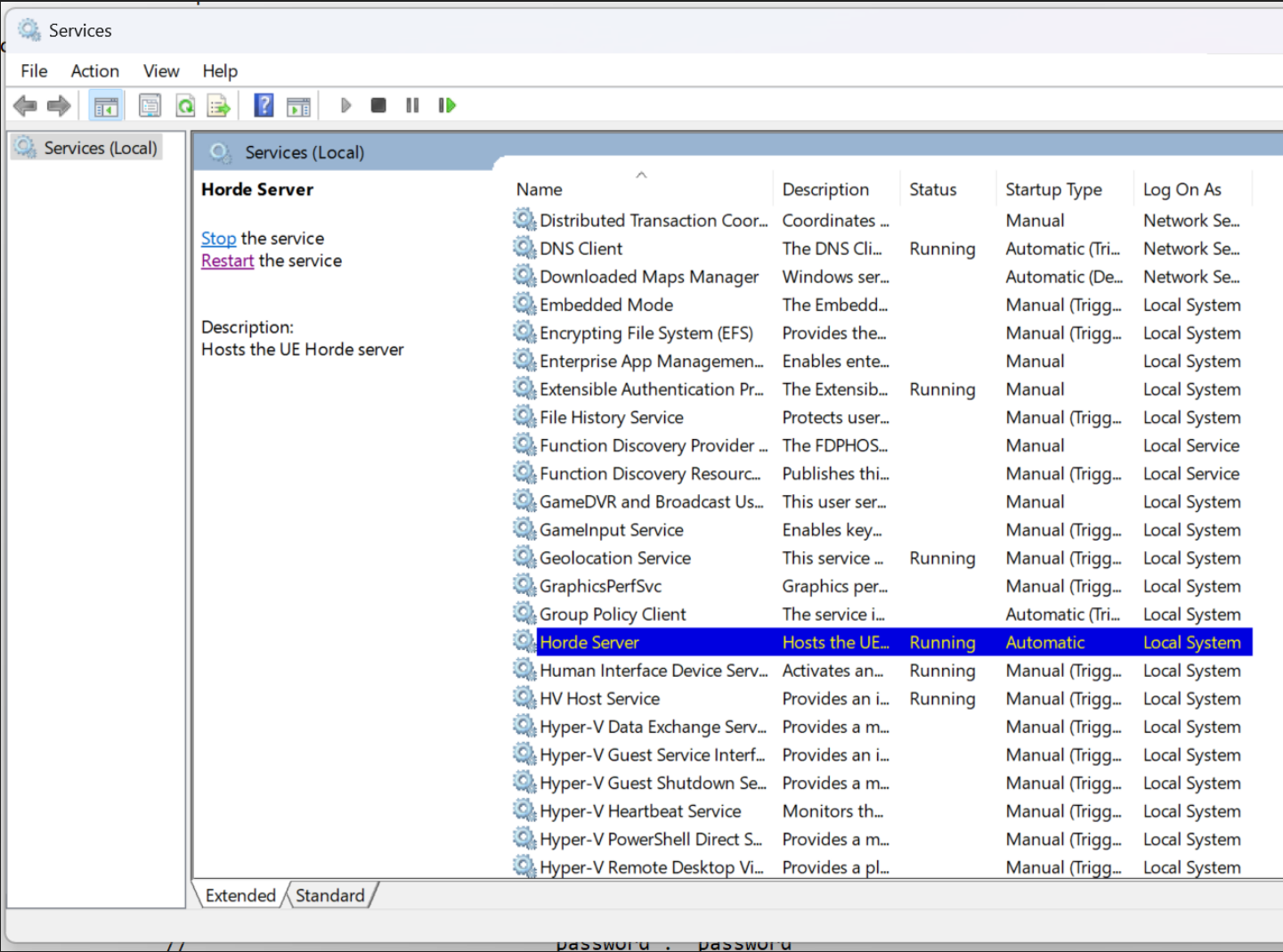
Where you'll find the `HttpPort` , `HttpPort2` , and where you could create the `HttpsPort` as DWORD (32-bit) Values.

Step 4.

Now that we've tweaked the `server.json` , we'll need to restart the server for it to load the new configuration. this is done by going to the Services control panel:



Finding the Horde Server service, and restarting it:



Step 5.

Now that we've rebooted the server with authentication enabled, we can once again open the dashboard at `http://localhost:13340` (or other IP/port combination that you configured), where you'll be prompted to set a password for the Admin user:



UNREAL ENGINE

HORDE

Username

Admin

Set Password

Optional

Confirm Password

Login

Step 6.

With the `server.json` configuration out of the way, we'll only have to configure the `global.json` file (also found in the `C:/ProgramData/Epic/Horde/Server` directory, next to `server.json`), which can be hot-reloaded by the server, and therefore does not need any reboots. Since it is hot-reloadable, the server also monitors for any configuration problems, and will notify you if it detects any (syntax) errors on its status page, found under `Server > Status` :

 HORDE

TOOLS

SERVER

HELP



Home >

Status

Service	Status	Time	Message
 GlobalConfig	Healthy	Dec 2nd (Today) at 2:13 AM PST	Operating normally.
 MongoDB	Healthy	Dec 2nd (Today) at 2:42 AM PST	Operating normally.
 Perforce	Healthy	Dec 2nd (Today) at 2:41 AM PST	Operating normally, see server status .
 Redis	Healthy	Dec 2nd (Today) at 2:42 AM PST	Operating normally.

The configuration for `globals.json` is shown below. In this config, once again a large part has been commented out and left for easy further configuration. The main part of the configuration is around the compute and analytics plugin configurations:

Tweak the file according to your requirements (see comments for details).

If using Horde authentication, make sure that any non administrator users that are created also have appropriate permissions where relevant. An agent doesn't need `AddComputeTasks` for helping with a task, but for initiating a new task it is needed.


```

    },
    "parameters": {
      "ugs": {
        // Set the default Perforce server for anyone using UGS.
        "defaultPerforceServer": "ssl:perforce.example.com:1666"
      }
    }
  },
  // Default deployment parameters for other tools bundled with Horde. The
  // "parameters": {
  //   "ugs": {
  //     // Set the default Perforce server for anyone using UGS.
  //     "defaultPerforceServer": "ssl:perforce.example.com:1666"
  //   }
  // }
}

```

globals.json

Step 7.

After making these tweaks, there is another tweak that I would recommend to the `default.global.json` that is loaded at the top of the `globals.json` file. this file can be found in the installation directory at `C:/Program Files/Epic Games/Horde/Server/Defaults`. The part we'll be tweaking is the pools definitions. By default the pools are configured like this:

```

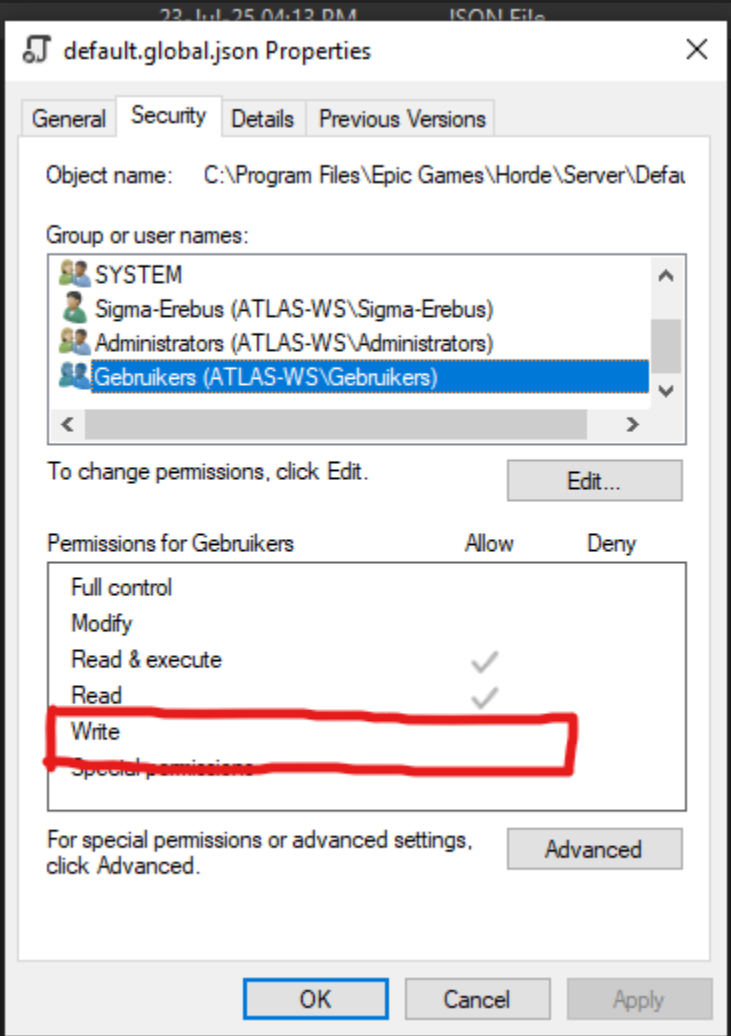
"pools": [
  {
    "name": "Win-UE5",
    "condition": "Platform == 'Win64'",
    "color": "Blue"
  },
  {
    "name": "Win-UE5-GPU",
    "condition": "Platform == 'Win64' && Interactive == true",
    "color": "Blue"
  },
  {
    "name": "Mac-UE5",
    "condition": "Platform == 'Mac'",
    "color": "Green"
  },
  {
    "name": "Linux-UE5",
    "condition": "Platform == 'Linux'",
    "color": "Orange"
  }
]

```

pool definitions within "plugins" > "compute"

These pools will be assigned automatically based on the capabilities of the agent that connects. These pools are also the things targeted when initiating a compile. So when starting a compile to the `win-UE5` pool, Linux agent machines that are capable of running Windows compiles through Wine will not be utilized. To fix this, I recommend rewriting the pool definitions to this, I have attached the full `default.global.json` below for easy copy-paste.

Note: since this is within Program Files, you might need additional permissions to write to files, to do so, open the file's properties on the Security tab, and make sure you have Write permission to the file:



Gebruikers = Users (used to be a dutch laptop)

```
"pools": [  
  {  
    "name": "Win-UE5",  
    "condition": "Platform == 'Win64'",  
    "color": "Blue"  
  },  
  {  
    "name": "Win-UE5-GPU",  
    "condition": "Platform == 'Win64' && Interactive == true",  
    "color": "Blue"  
  },  
  {  
    "name": "Win-UE5-Compatible",  
    "condition": "Platform == 'Win64' || WineEnabled == true",  
    "color": "Blue"  
  },  
  {  
    "name": "Mac-UE5",  
    "condition": "Platform == 'Mac'",  
    "color": "Green"  
  },  
  {  
    "name": "Linux-UE5",  
    "condition": "Platform == 'Linux'",  
    "color": "Orange"  
  },  
  {  
    "name": "Linux-UE5-Wine",  
    "condition": "Platform == 'Linux' && WineEnabled == true",  
    "color": "Orange"  
  }  
]  
]
```

altered pool definitions within "plugins" > "compute"

```
{
  // Settings for the dashboard
  "dashboard": {

    // Whether to show the default landing page with documentation when
    "showLandingPage": true,

    "showCI": true,
    "showAgents": true,
    "showPerforceServers": true,
    "showDeviceManager": true,
    "showTests": true
  },

  // Settings for various plugins
  "plugins": {

    "compute": {
      "clusters": [
        //{
        //  "id": "default",
        //  "namespaceId": "horde.compute"
        //}
      ],
      "networks": [
        {
          "id": "default",
          "cidrBlock": "0.0.0.0/0",
          "description": "Default network (matching all IPs)",
          "computeId": "default"
        }
      ],
      "pools": [
        {
          "name": "Win-UE5",
          "condition": "Platform == 'Win64'",
          "color": "Blue"
        },
        {
          "name": "Win-UE5-GPU",
          "condition": "Platform == 'Win64' && Interactive == true",
          "color": "Blue"
        },
        {
          "name": "Win-UE5-Compatible",
          "condition": "Platform == 'Win64' || WineEnabled == true",
          "color": "Blue"
        },
        {
          "name": "Mac-UE5",
          "condition": "Platform == 'Mac'",
          "color": "Orange"
        },
        {
          "name": "Linux-UE5",
          "condition": "Platform == 'Linux'",
          "color": "Green"
        },
        {
          "name": "Linux-UE5-Wine",
          "condition": "Platform == 'Linux' && WineEnabled == true",
          "color": "Green"
        }
      ]
    }
  }
}
```

```
    },
  ],
},

"storage": {
  "backends": [
    {
      "id": "default-backend",
      "type": "FileSystem",
      "baseDir": "Storage" // Default base directory is C:\ProgramData\Horde\
    },
    {
      "id": "memory-backend", // Used for automated tests
      "type": "Memory"
    }
  ],
  "namespaces": [
    {
      "id": "default",
      "backend": "default-backend"
    },
    {
      "id": "horde-artifacts",
      "prefix": "Artifacts/",
      "backend": "default-backend"
    },
    {
      "id": "horde-perforce",
      "prefix": "Perforce/",
      "backend": "default-backend"
    },
    // The 'horde-logs' namespace is used internally by Horde to
    // and remote execution systems. It must be configured in order
    // correctly.
    {
      "id": "horde-logs",
      "prefix": "Logs/",
      "backend": "default-backend"
    },
    {
      "id": "horde-tools",
      "prefix": "Tools/",
      "backend": "default-backend"
    },
    {
      "id": "memory",
      "backend": "memory-backend",
      "enableAliases": true
    }
  ]
},

"build": {
  "artifactTypes": [
    {
      "name": "step-output",
      "keepDays": 1
    }
  ],
  "devices": {
    "platforms": [
      {
        "id": "android",
        "name": "Android"
```

```

    }
  ],
  "pools": [
    {
      "id": "ue5",
      "name": "UE5",
      "poolType": "Automation"
    },
    {
      "id": "remote-ue5",
      "name": "Remote UE5",
      "poolType": "Shared"
    }
  ]
}
},
// Default ACL groups
"acl": {
  "entries": [
    {
      "claim": {
        "type": "http://epicgames.com/ue/horde/group",
        "value": "View"
      },
      "profiles": [
        "default-read"
      ]
    },
    {
      "claim": {
        "type": "http://epicgames.com/ue/horde/group",
        "value": "Run"
      },
      "profiles": [
        "default-run"
      ]
    }
  ]
}
}

```

full default.global.json

These conditions can also be customized to evaluate custom properties in an agent's configuration to pool agents in more nuanced ways. These custom properties can be defined in the `agent.json` file (details will be in agent setup guides). But once configured, the conditions can also evaluate based on those custom keys to have more compute pools:

```

{
  "name": "Team-A",
  "condition": "Team == 'Team-A'",
  "color": "Grey"
},

```

if desired, colors here can be tweaked as well (my personal preference is to swap around the colors of Mac & Linux). However for colors you can only pick from: Blue, Orange, Green, Grey and Default.

Step 8.

After all this setup, you might want or need to restart the server every now and then, which can be done by repeating Step 4.

Member discussion

0 comments

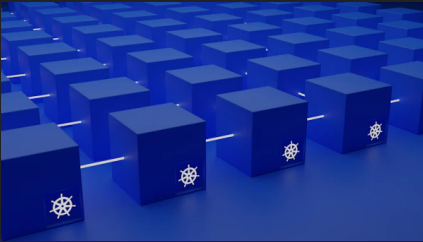
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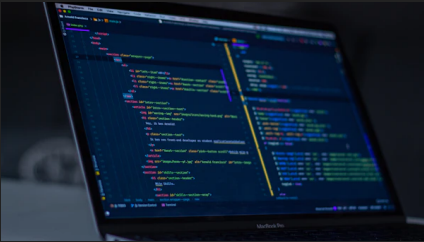
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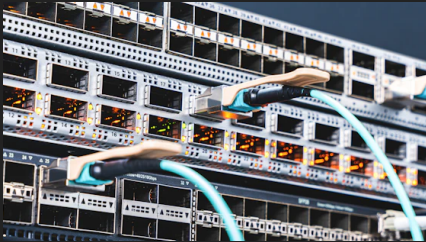
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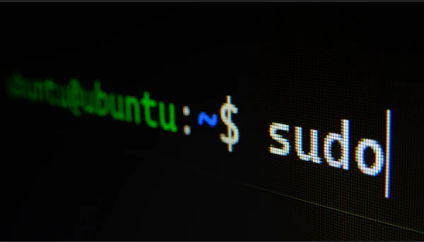
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